Prakash V

Software Developer | Full Stack | Python | React | Spring Boot
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Summary

Pre-final year B.Tech Information Technology student (graduating 2028) building production-style projects: React frontends, Spring Boot and FastAPI backends, and a self-hosted RAG system. 24 public repositories, 400+ automated tests, 200+ LeetCode problems. Clear communication and teamwork skills. Seeking an entry-level Software Developer role.

Technical Skills

Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, Bash, C++
Frontend: React, HTML, CSS, Tailwind CSS, Framer Motion, TensorFlow.js
Backend: Spring Boot, FastAPI, REST APIs, JWT Auth, PostgreSQL, MySQL, Redis, SQLite, ChromaDB
AI & ML: Machine Learning, RAG, AI/LLM integrations (Gemini LLM APIs), sentence-transformers, COCO-SSD
DevOps & Cloud: Git, GitHub Actions (cloud CI/CD), Docker, Linux, Postman
Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems
Practices: Unit Testing (pytest), CI/CD, Agile, Scrum, Code Reviews, Debugging, Version Control, Problem Solving

Experience

Independent Software Developer - Personal & Open-Source Projects *Sep 2024 - Present*

- Delivered 6 production-style applications and 24 public repositories across React, Spring Boot, and FastAPI, backed by 400+ unit and integration tests and code-review checklists on GitHub Actions CI.
- Collaborated with peers on college group projects and open-source work using Git workflows, Agile and Scrum practices, and clear written and verbal communication.

Projects

QA-Assistant "Marginalia" - Self-Hosted RAG Document Q&A *Jun 2026 - Sep 2026*
Tech: Python, FastAPI, React, ChromaDB, Gemini API, Docker, GitHub Actions

- Built a privacy-first RAG application that answers questions over 3 file formats (PDF, DOCX, TXT) with per-claim source citations, confidence scores, and streamed responses.
- Added semantic chunking, BGE cross-encoder reranking, local embeddings, and PII / prompt-injection guardrails; maintained 325 pytest tests with a RAGAS evaluation gate in CI.

TaskFlow - Background Job Processing Platform *Jul 2026 - Sep 2026*
Tech: Java 21, Spring Boot, Redis, PostgreSQL, React 19, Docker Compose

- Designed an asynchronous job platform with REST enqueue, Redis queue, and a headless worker featuring exponential-backoff retries and a dead-letter queue for monitoring failed jobs.
- Guaranteed exactly-once execution with atomic conditional UPDATE claims (12 jobs across 3 worker replicas, 0 duplicates); 76 JUnit tests, cursor pagination, Flyway migrations.

Fleet Management & GPS Vehicle Tracking System *Apr 2026 - Jun 2026*
Tech: Spring Boot 3.5, Spring Security (JWT), React 19, MySQL, Leaflet

- Built a full-stack fleet system with live GPS tracking on interactive maps, geofence entry/exit alerts, and telemetry events for speeding, idling, and harsh braking.
- Secured 11 REST controllers with stateless JWT authentication and role-based access control for 6 roles; automated trip-compliance checks across documents and maintenance.

AI Vision - Edge Object Detection Platform *Jan 2026 - Feb 2026*
Tech: React 19, Vite, TensorFlow.js, COCO-SSD, WebGL, Framer Motion

- Delivered in-browser detection of 80+ object classes with WebGL-accelerated, sub-100ms inference and real-time canvas bounding boxes, all processing on-device with zero image uploads.

Education

Bachelor of Technology (B.Tech), Information Technology *Sep 2024 - Expected 2028*
Sri Krishna College of Technology, Coimbatore - CGPA: 7.58/10

Certifications & Achievements

- **freeCodeCamp - Responsive Web Design Certification** (HTML, CSS, JavaScript)
- **LeetCode - 200+ problems solved** in data structures and algorithms

Languages: English, Hindi